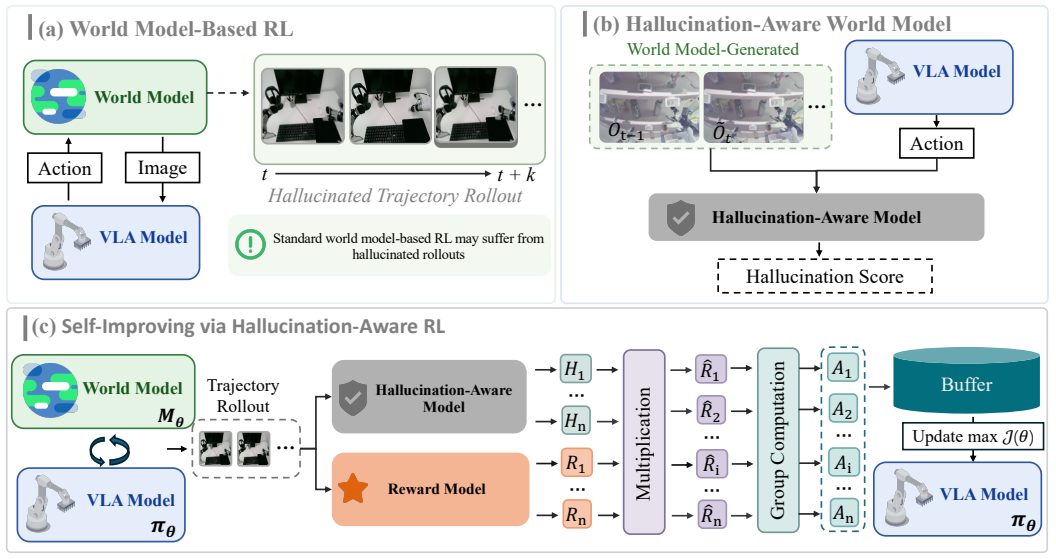

(a) World Model-Based RL
World Model
Action
Image
VLA Model
t
t + k
Hallucinated Trajectory Rollout
Standard world model-based RL may suffer from hallucinated rollouts
(b) Hallucination-Aware World Model
World Model-Generated
VLA Model
Action
Hallucination-Aware Model
Hallucination Score
(c) Self-Improving via Hallucination-Aware RL
World Model
VLA Model
Trajectory Rollout
Hallucination-Aware Model
Reward Model
Multiplication
Group Computation
Buffer
Update max
VLA Model